21 A

Equating $I = nqv_d A$ and $I = \frac{V}{R} = \frac{V}{\rho L} A$,

$$v_d = \frac{V}{\rho L n q} \propto \frac{1}{L}$$

We need to find the new length of the conducting putty.

The volume is conserved, so $AL = (\frac{A}{4})L' \rightarrow L' = 4L$

Hence $\frac{v_d'}{v_d} = \frac{L}{L'} = \frac{1}{4}$

22 C Total energy that can be supplied by the battery E = (12 V)(60 Ah)